Team Number:	105
Project Name:	**Radiolink**
Team Member Names:	**Joseph Rivera, Mateo Delgado, Sultan Haidar Ali, William Buckner**
Version:	**#1**

A. List ALL major components (active devices, integrated circuits, etc.) exc

All Major Components	Component Name
	Temperature Sensor
	Operational Amplifier
	PIC18F57Q43 Curiosity Nano

B. Assign each major component above to ONE power rail below. Try to m

+5V Power Rail	Component Name
	Temperature Sensor
	Operational Amplifier
	PIC18F57Q43 Curiosity Nano

C2. Regulator or Source Choice	+5V Regulator

C. For each power rail above, select a specific voltage regulator using the s

D. Select a specific external power source (wall supply or battery) for your

External Power Source 1	Component Name
Power Source 1 Selection	Plug-in Wall Supply
Power Rails Connected to External Power Source 1	
	+5V Regulator

E. Calculate Battery Life (if applicable). For each battery, also check the w

	Component Name

N/A

Notes

Rivera Power Budget

cept for power sources, voltage regulators, resistors, capacitors, or passi

Part Number	SupplyVoltageRange	#
MCP9700A-E/TO	2.3 - 5.5 V	1
MCP6004-I/SL	1.8 - 6.0 V	1
DM164150	1.8 - 5.5 V	1

inimize the number of different power rails in the design.

Part Number	SupplyVoltageRange	#
MCP9700A-E/TO	2.3 - 5.5 V	1
MCP6004-I/SL	1.8 - 6.0 V	1
DM164150	1.8 - 5.5 V	1

LM7805	5.0 - 9.0 V	1

Tota

same process as for major component selection. Confirm that the Total R

system, and confirm that it can supply all of the regulators for all of the p

Part Number	SupplyVoltageRange	Output
AlloverPower B07QV8CCJ5	100 - 240 VAC	9
LM7805	7 - 25 VAC	1

Total Remaining Curr

orst-case lifetime of the battery by indicating the capacity in mAh.

Part Number	SupplyVoltageRange

N/A N/A

ive elements		
AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
0.012	0.012	mA
0.68	0.68	mA
25	25	mA
	0	mA
	0	mA
	25.692	mA

AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
0.012	0.012	mA
0.68	0.68	mA
25	25	mA
	0	mA
	0	mA
Subtotal	25.692	mA
Safety Margin	25%	
Total Current Required on +5V Rail	32.115	mA
1500	1500	mA
l Remaining Current Available on +5V Rail	1467.885	mA

Remaining Current Available on each rail above is not negative.

power rails simultaneously. If you need multiple power sources, list each

AbsoluteMaximumCurrent (mA)	TotalCurrent(mA)	Unit
3000	3000	mA
	0	mA
1500	1500	mA
	0	mA
rent Available on External Power Source 1	1500	mA

Capacity(mAh)	RequiredByRegulators	

N/A		N/A	
	Battery Life	N/A	hours